

COURSE SYLLABUS

STAT452 – Applied Statistics for Engineering and Scientists II

1. GENERAL INFORMATION

Course name:	Applied Statistics for Engineering and Scientists II
Course name (in Vietnamese):	Thống kê ứng dụng trong kỹ thuật và khoa học 2
Course ID:	STAT452
Knowledge block:	Elective - Math
Number of credits:	4
Credit hours for theory:	40
Credit hours for practice:	30
Credit hours for self-study:	90
Pre requisite:	
Prior-course:	STAT451
Instructors:	Dr. Hoàng văn Hà, Dr. Nguyễn Thị Mộng Ngọc

2. COURSE DESCRIPTION

This course provides an in-depth understanding of applied statistical methods used in engineering and scientific research, with a strong focus on regression analysis and experimental design. Students will explore multiple linear regression, polynomial and logistic regression, as well as variable selection techniques such as Ridge and LASSO. The course also introduces the design of experiments, including one-factor and two-factor ANOVA, and covers key statistical tools such as goodness-of-fit tests. Through case studies, students apply statistical techniques to real-world problems and datasets. A key component of the course is the use of R programming for data analysis, visualization, and interpretation, equipping students with practical skills in statistical computing.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Work individually and collaboratively to complete statistical projects and case studies	LO6, LO9, LO11

ID	Description	Program LOs
G2	Interpret statistical problems and results using correct terminology in English	LO10
G3	Explain fundamental concepts of regression and experimental design	LO1, LO12
G4	Apply statistical models to analyze engineering and scientific data	LO1, LO2, LO3
G5	Use R programming language to perform data visualization and statistical data analysis.	LO4, LO5, LO12

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Organize, operate and manage teamwork effectively through case studies and project-based assignments	I
G1.2	Propose statistical approaches and discuss results logically in class and group work	IT
G1.3	Present statistical findings clearly in oral presentations and written reports	TU
G2.1	Use the skill of reading and interpreting statistical documentation and results in English	T
G3.1	Explain fundamental concepts of regression, hypothesis testing, and experimental design	IT
G4.1	Apply knowledge of statistics and computing in analyzing engineering and scientific problems	IT
G4.2	Apply statistical models and programming foundations to solve data-driven problems	IT
G4.3	Analyze real-world datasets and define suitable statistical methods and tools for proper data analysis	ITU
G5.1	Use R programming to perform statistical analysis and create data visualizations	U

5. TEACHING PLAN

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)
1	Mutiple linear regression	G2.1, G3.1, G4.1, G5.1	Lecturing Q&A, Group discussion
2	Polynomial regression and transformation	G2.1, G3.1, G4.1, G5.1	Lecturing Demonstration, Q&A
3	Ridge and LASSO regression – Variable selection	G2.1, G3.1, G4.1, G5.1	Lecturing Demonstration, discussion
4	Case study on Chapter 1 to Chapter 3	G1.1, G1.2, G4.2, G4.3, G5.1	Lecturing Demonstration, discussion
5	Logistic regression	G2.1, G3.1, G4.1, G5.1	Lecturing Demonstration
6	Midterm		
7	Introduction to projects. Design of experiment: one-factor ANOVA	G2.1, G3.1, G4.1, G5.1, G4.2	Lecturing Demonstration
8	Design of experiment: two-factor ANOVA	G2.1, G3.1, G4.1, G5.1, G4.2	Lecturing Q&A, discussion
9	Case study on Chapter 7 to Chapter 8.	G1.1, G1.2, G4.2, G4.3, G5.1	Lecturing
10	Goodness-of-fit Tests	G2.1, G3.1, G4.1, G5.1, G4.2	Lecturing Q&A, discussion
11	Project presentation	G1.1, G1.2, G1.3, G4.2, G4.3	Presentation

6. LABORATORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Mutiple linear regression	G2.1, G3.1, G4.1, G5.1	Lecturing Demonstration	
2	Polynomial regression and transformation	G2.1, G3.1, G4.1, G5.1	Lecturing Demonstration	
3	Ridge and LASSO regression – Variable selection	G2.1, G3.1, G4.1, G5.1	Lecturing Demonstration	
4	Review, Exercises	G1.1, G1.2, G1.3, G2.1	Lecturing Demonstration	
5	Logistic regression	G2.1, G3.1, G4.1, G5.1, G4.2	Lecturing Demonstration	
6	Review, Exercises	G1.1, G1.2, G1.3, G2.1	Lecturing Demonstration	
7	Design of experiment: one-factor ANOVA	G2.1, G3.1, G4.1, G5.1, G4.2	Lecturing Demonstration	
8	Design of experiment: two-factor ANOVA	G2.1, G3.1, G4.1, G5.1, G4.2	Lecturing Demonstration	
9	Review, Exercises	G1.1, G1.2, G1.3, G2.1	Lecturing Demonstration	
10	Goodness-of-fit Tests	G2.1, G3.1, G4.1, G5.1, G4.2	Lecturing Demonstration	

7. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Homework Assignments		G1.1, G1.2, G1.3, G2.1, G3.1, G4.1, G4.2, G4.3, G5.1	25%
A2	Exams			75%
A21	Midterm exam	Closed book exam.	G2.1, G3.1, G4.1, G4.2, G4.3	35%

A22	Final project		G2.1, G3.1, G4.1, G4.2, G4.3	40%
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8. RESOURCES

Textbooks

- J. Devore, Brooks/Cole, *Probability and Statistics for Engineering and the Sciences*, Cengage Learning, 9th, 2016.
- Montgomery, Douglas C. *Design and analysis of experiments*. John wiley & sons, 2017.

Others

- Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, *An Introduction to Statistical Learning: with Applications in R*, Springer, 2015.
- Sanford Weisberg, *Applied Linear Regression*, 4th Edition, Wiley, 2014.
- Douglas C. Montgomery, Elizabeth A. Peck & G. Geoffrey Vining, *Introduction to Linear Regression Analysis*, 4th Edition, Wiley, 2012.
- Jared P. Lander, *R for Everyone: Advanced Analytics and Graphics*, 2nd Edition, Addison-Wesley, 2017.
- R. E. Walpole, R. H. Myers, S. L. Myers, and K. Ye, *Essentials of Probability & Statistics for Engineers & Scientists*, Pearson, 2013.

Software

- R

9. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.